

Dealing with Outliers

In practice, although an extreme value or outlier in a financial dataset may represent a rare value in the population, it may also reflect an error in recording the value of an observation or an observation generated from a different population. After having checked and eliminated errors, we can address what to do with extreme values in the sample.

When dealing with a sample that has extreme values, there may be a possibility of transforming the variable (e.g., a log transformation) or of selecting another variable that achieves the same purpose. If, however, alternative model specifications or variable transformations are not possible, then three options exist for dealing with extreme values:

Option 1 Do nothing; use the data without any adjustment.

Option 2 Delete all the outliers.

Option 3 Replace the outliers with another value.

The first option is appropriate if the values are legitimate, correct observations, and it is important to reflect the whole of the sample distribution. Because outliers may contain meaningful information, excluding or altering these values may reduce valuable information. Further, because identifying a data point as extreme leaves it up to the judgment of the analyst, leaving in all observations eliminates that need to judge a value as extreme.

The second option excludes the extreme observations. One measure of central tendency in this case is the **trimmed mean**, which is computing an arithmetic mean after excluding a stated small percentage of the lowest and highest values. For example, a 5 percent trimmed mean discards the lowest 2.5 percent and the highest 2.5 percent of values and computes the mean of the remaining 95 percent of values. A trimmed mean is used in sports competitions when judges' lowest and highest scores are discarded in computing a contestant's score.

The third option involves substituting values for the extreme values. A measure of central tendency in this case is the **winsorized mean**. It is calculated after assigning one specified low value to a stated percentage of the lowest values in the dataset and one specified high value to a stated percentage of the highest values in the dataset. For example, a 95 percent winsorized mean sets the bottom 2.5 percent of values in the dataset equal to the value at or below which 2.5 percent of all the values lie (as will be seen shortly, this is called the "2.5th percentile" value) and the top 2.5 percent of values in the dataset equal to the value at or below which 97.5 percent of all the values lie (the "97.5th percentile" value).

Often comparing the statistical measures of datasets with outliers included and with outliers excluded can reveal important insights about the dataset. Such comparison can be particularly helpful when investors analyze the behavior of asset returns and rate, price, spread and volume changes.

In Example 1, we show the differences among these options for handling outliers using daily returns for the fictitious Euro-Asia-Africa (EAA) Equity Index in Exhibit 2.

EXAMPLE 1

Handling Outliers: Daily Returns to an Index

Using daily returns on the EAA Equity Index for the past five years, consisting of 1,258 trading days, we can see the effect of trimming and winsorizing the data: